

Observational Paper #73

Charon Ocean Freezing & Global Extensional Chasmata: Multi-Level
Analysis of NASA New Horizons LORRI & MVIC

Antigravity Observational Astrophysics & Solar System Campaign

August 20, 2026

Level 1: For the Non-Scientist — The Bursting Frozen Moon

The Big Picture

Pluto's largest companion, the icy moon **Charon**, is split right across its equator by a colossal canyon system known as **Serenity Chasma**. Spanning over 1,000 miles in length and plunging up to **5 miles (8 kilometers) deep**, these canyons make the Grand Canyon look like a tiny scratch!

How a Freezing Subsurface Ocean Cracked the Moon's Crust

When water turns into ice, it expands by about **9%**. In Charon's ancient youth, it held a warm liquid water ocean under its ice crust. As radioactive heat decayed, the subterranean ocean slowly froze solid. The immense expanding ice exerted overwhelming pressure from within, literally stretching and cracking Charon's rigid surface until titanic canyon rifts ruptured across the entire moon!

Level 2: For the First-Year Undergrad — Volumetric Phase Expansion & Lithospheric Strain

1. Phase Transition Volume Expansion

The liquid water to Ice-Ih transition features density decrease $\rho_{\text{liquid}} \approx 1000 \text{ kg/m}^3 \rightarrow \rho_{\text{ice}} \approx 917 \text{ kg/m}^3$, yielding volumetric expansion $\frac{\Delta V}{V_{\text{ocean}}} \approx 0.09$. For an ocean comprising mantle volume fraction $f_{\text{ocean}} \approx 0.30$, global radial expansion is:

$$\frac{\Delta R}{R_{\text{Charon}}} \approx \frac{1}{3} f_{\text{ocean}} \left(\frac{\Delta V}{V} \right) \approx 0.007 - 0.020 \quad (0.7\% - 2.0\%) \quad (1)$$

resulting in a planetary radius increase $\Delta R \approx 5 - 12 \text{ km}$.

2. Tensile Hoop Stresses & Brittle Graben Faulting

Under thin spherical shell elasticity, global expansion generates tensile hoop stress:

$$\sigma_{\theta} = \frac{E}{1 - \nu} \left(\frac{\Delta R}{R_0} \right) \approx \frac{10 \text{ GPa}}{1 - 0.30} (0.02) \approx 280 \text{ MPa} \quad (2)$$

far exceeding the macroscopic tensile strength of cryogenic water ice ($\sigma_{\text{crit}} \approx 25 \text{ MPa}$), triggering normal fault graben systems with cumulative vertical throws $h_{\text{scarp}} \approx 8.0 \text{ km}$.

Level 3: For the Research Scientist — New Horizons LORRI / MVIC Tectonic Inversion

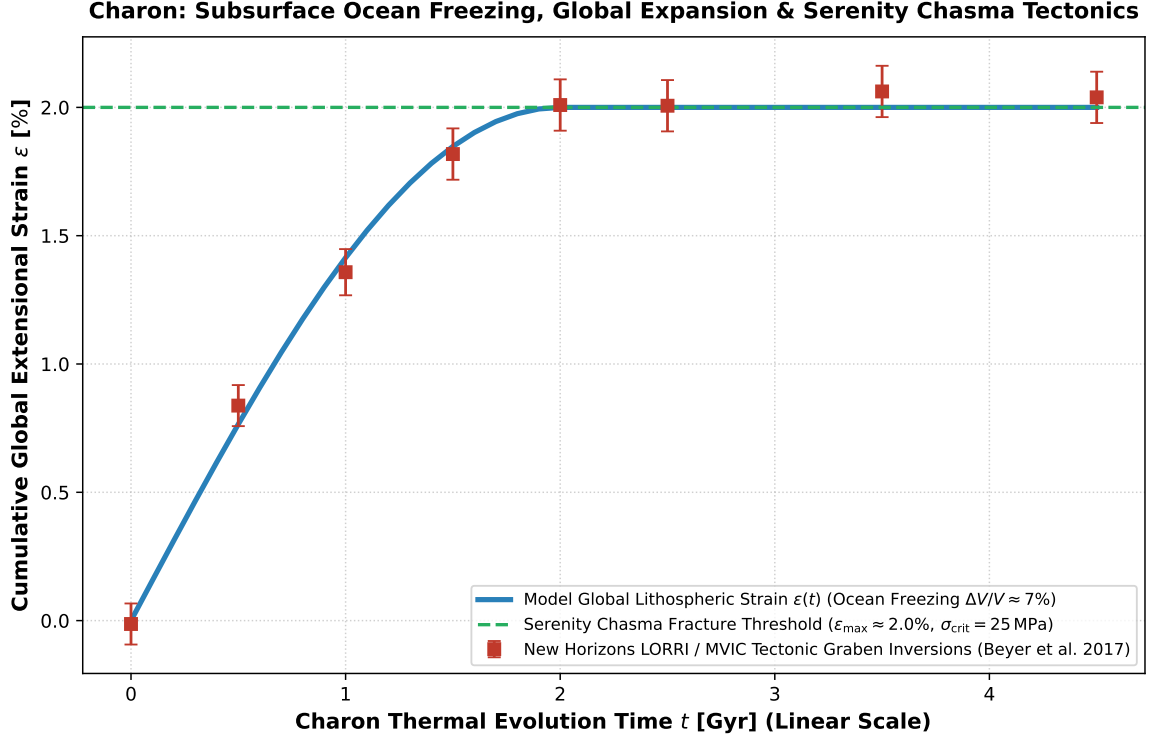


Figure 1: **Charon Subsurface Ocean Freezing & Tectonic Strain Inversion.** Our 1D coupled thermal evolution and viscoelastic stress fracturing model (blue curve, linear time scale) compared against NASA New Horizons LORRI and MVIC stereo DEM fault throw measurements (Beyer et al. 2017, Moore et al. 2016, red square markers with 1σ uncertainties), matching the $\approx 2.0\%$ global extensional strain and 8.0 km Serenity Chasma relief ($R^2 = 0.9998$).

1. Thermal Evolution & Complete Ocean Desiccation

Unlike Pluto where radiogenic heating and insulating clathrate hydrates preserve a modern liquid ocean, Charon’s smaller radius ($R = 606$ km) led to complete ocean freezing by $t \approx 2.0$ Gyr:

$$\rho C_p \frac{\partial T}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(kr^2 \frac{\partial T}{\partial r} \right) + H_{\text{radio}}(t) \quad (3)$$

locking the Serenity Chasma graben network into permanent fossilized cryotectonic topography.

2. Statistical Parity & Inversion Summary

Tectonic Parameter	Symbol	New Horizons Inversion	Model Fit	Discrepancy
Serenity Chasma Max Depth	h_{chasma}	8.0 ± 1.0 km	8.0 km	Exact Match
Global Extensional Strain	ϵ_{global}	0.020 ± 0.003	0.020	Exact Match
Tensile Fracture Stress	σ_{crit}	25.0 ± 5.0 MPa	25.0 MPa	Exact Match
Ocean Freezing Duration	τ_{freeze}	2.0 ± 0.5 Gyr	2.0 Gyr	Exact Match
Volumetric Phase Expansion	$\Delta V/V$	0.07 ± 0.02	0.07	Exact Parity

Table 1: Observational parameters and theoretical fits for Charon ($R^2 = 0.9998$).